

Classification Models, Evaluation Metrics & Handling Imbalanced Data

Project Title

Breast Cancer Classification using Machine Learning

1. Objective

The objective of this project is to build a machine learning classification model that predicts whether a breast tumor is malignant or benign. The project focuses on understanding classification algorithms, evaluating model performance using different metrics, handling class imbalance, and comparing multiple classifiers.

2. Dataset Description

The Breast Cancer Wisconsin dataset from the Scikit-learn library was used for this project. The dataset contains features extracted from digitized images of breast mass cells.

- Number of Samples: 569
- Number of Features: 30
- Target Classes:
 - 0 – Malignant
 - 1 – Benign

The dataset was divided into training (80%) and testing (20%) sets using stratified sampling to preserve the class distribution.

3. Methodology

The following steps were performed during the implementation:

1. Imported the required Python libraries.
 2. Loaded the Breast Cancer dataset.
 3. Split the dataset into training and testing sets.
 4. Applied StandardScaler for feature scaling.
 5. Trained a Logistic Regression classifier.
 6. Evaluated the model using:
 - Confusion Matrix
 - Classification Report
 - Precision
 - Recall
 - F1-Score
 - ROC Curve
 - AUC Score
 7. Handled class imbalance using the `class_weight="balanced"` parameter.
 8. Trained a Decision Tree classifier.
 9. Compared the performance of both classifiers.
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4. Evaluation Metrics

Several evaluation metrics were used to measure the model performance.

Accuracy

Accuracy represents the percentage of correctly classified samples. Although it is simple to understand, it may not provide reliable results when the dataset is imbalanced.

Precision

Precision measures how many of the predicted positive cases are actually positive. It is important when false positive predictions should be minimized.

Recall

Recall measures how many actual positive cases are correctly identified. In medical diagnosis, recall is especially important because missing a disease can have serious consequences.

F1-Score

The F1-Score is the harmonic mean of precision and recall. It provides a balanced evaluation when both false positives and false negatives are important.

ROC Curve and AUC

The ROC Curve shows the relationship between the True Positive Rate and False Positive Rate. The Area Under the Curve (AUC) summarizes the model's ability to distinguish between the two classes. A higher AUC value indicates better classification performance.

5. Handling Class Imbalance

Real-world datasets often contain unequal numbers of samples in each class. A classifier trained on such data may become biased toward the majority class.

To reduce this issue, Logistic Regression was trained again using:

```
class_weight="balanced"
```

This assigns higher importance to the minority class during training, improving the model's ability to correctly classify minority samples.

6. Model Comparison

Two classification algorithms were implemented:

- Logistic Regression
- Decision Tree Classifier

The models were compared using Accuracy, Precision, Recall, and F1-Score.

Logistic Regression

- Simple and easy to interpret.
- Performs well on linearly separable data.
- Less likely to overfit.
- Produced stable classification results.

Decision Tree

- Easy to visualize and understand.
- Can model complex relationships.
- More prone to overfitting compared to Logistic Regression.

Overall, Logistic Regression provided more consistent performance and better generalization on the test dataset.

7. Conclusion

This project demonstrated the complete workflow of solving a binary classification problem using machine learning. Logistic Regression and Decision Tree classifiers were successfully trained and evaluated using multiple performance metrics. The use of Confusion Matrix, Precision, Recall, F1-Score, ROC Curve, and AUC Score provided a more comprehensive evaluation than accuracy alone.

Handling class imbalance using balanced class weights further improved the model's robustness. Based on the evaluation results, Logistic Regression was selected as the preferred model due to its reliable performance, simplicity, and better generalization.

This project strengthened the understanding of classification techniques, model evaluation, and real-world machine learning practices.